

Evaluation of AI content generation tools for verification of academic integrity in higher education

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Abstract

Purpose – The notion of using a generative artificial intelligence (AI) engine for text composition has gained excessive popularity among students, educators and researchers, following the introduction of ChatGPT. However, this has added another dimension to the daunting task of verifying originality in academic writing. Consequently, the market for detecting artificially generated content has seen a mushroom growth of tools that claim to be more than 90% accurate in sensing artificially written content.

Design/methodology/approach – This research evaluates the capabilities of some highly mentioned AI detection tools to separate reality from their hyperbolic claims. For this purpose, eight AI engines have been tested on four different types of data, which cover the different ways of using ChatGPT. These types are Original, Paraphrased by AI, 100% AI generated and 100% AI generated with Contextual Information. The AI index recorded by these tools against the datasets was evaluated as an indicator of their performance.

Findings – The resulting figures of cumulative mean validate that these tools excel at identifying human generated content (1.71% AI content) and perform reasonably well in labelling AI generated content (76.85% AI content). However, they are perplexed by the scenarios where the content is either paraphrased by the AI (39.42% AI content) or generated by giving a precise context for the output (60.1% AI content).

Originality/value – This paper evaluates different services for the detection of AI-generated content to verify academic integrity in research work and higher education and provides new insights into their performance.

Keywords Chatbot, Generative AI, ChatGPT, LLM, AI content detectors, AI writing

Paper type Technical paper

Introduction

A long evolution in the field of language models led to the emergence of the Generative Pre-trained Transformer (GPT) (Floridi and Chiriatti, 2020). The birth of first language models in the 1960s (Weizenbaum, 1966) and advancements which occurred with the invention of novel deep learning techniques, such as transformer in 2017 (Vaswani *et al.*, 2017), led to the revolutionary breakthrough of ChatGPT (an AI chatbot powered by GPT3.5) in November 2022. It was introduced by the OpenAI platform and earned more popularity than any other AI text generation engine due to its ability to produce human-like responses by taking the context of the conversation into account.

The arrival of ChatGPT imparted a huge impact in miscellaneous domains (Haleem *et al.*, 2022). Researchers have not only found it useful in music and literature synthesis (Oksanen *et al.*, 2023; Perez-Castro *et al.*, 2023) but also extracted its benefits in healthcare and diagnostics (Antaki *et al.*, 2023; Castonguay *et al.*, 2023; Xiao *et al.*, 2023). It has also been reported to be an efficient tool for management and planning (Keiper, 2023; Keiper *et al.*,



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2023). However, within a few months of its introduction, a plethora of discussion surfaced about its ethical use from an educational perspective.

In this regard (Lo, 2023), reviewed the effectiveness of ChatGPT's usage in multiple teaching and learning domains during its initial months and highlighted potential integrity issues associated with its use in assessment methodologies. Though (Shoufan, 2023), listed down several positive reviews of using ChatGPT by students (Neumann *et al.*, 2023), talked about challenges associated with ChatGPT's deployment at university-level programmes and raised some important questions, such as how to check plagiarism of text created by ChatGPT, how to cite text produced by ChatGPT or what percentage of text generated by this chatbot was acceptable in a complete document?

On a similar note (Else, 2023), claimed that scientific reviewers could also get perplexed when it came to identifying the real text from the AI-generated text within research papers and further elaborated that commercial plagiarism detection software also marked the contents of AI generated text as 100% plagiarism-free. Since ChatGPT challenged the conventional educational paradigm, it made the academicians rethink about its use, explore different ways to tailor the conventional approach of education towards a new AI driven era and take steps to mitigate its destructive effects (Ahmed and Sharo, 2023; Dwivedi *et al.*, 2023; Yu, 2023).

ChatGPT produced false responses to the queries (Amaro *et al.*, 2023; Oviedo-Trespalacios *et al.*, 2023); however, the reinforcement learning technique at its back end improved the learning model with time (Wu *et al.*, 2023). To supplement this, intelligent contextual information in the form of prompts could also generate a better answer. In this context, (Spasić and Janković, 2023) discussed ways of successful prompt development to generate educational strategies for students' teaching and learning within the ChatGPT platform. This avoided fallacious and erroneous responses.

In addition to the above-mentioned perspectives of AI-generated content, (Currie *et al.*, 2023) acknowledged the efforts of commercial plagiarism software detection tools for providing revised versions of their products and services with an additional AI index. This index indicated the probability measure of AI-generated content. They specifically pointed towards the Turnitin release of April 2023. However (Dalalah and Dalalah, 2023), questioned the reliability of plagiarism detection tools by detecting false positives from the test cases produced before the inception of ChatGPT. Homolak (2023) tested a large set of the scientific abstracts on Alzheimer's disease (AD) from the SCOPUS database using a single publicly available AI content detector, and the results showed that 59% of the articles were truly composed by humans and the rest had AI content above 0%. However, for abstracts reported to have more than 0% AI-generated content, a threshold was not discussed to further distinguish the content as AI or human-generated.

A summary detailing the works of different researchers in this domain is presented in Table 1. The table also lists the pros and cons of the relevant works.

This work provides a relative examination of popular AI content detection tools, all of which provide some probabilistic measure of the content being generated through an AI platform, and tries to answer the following research questions (RQs).

RQ1. Are AI-generated content detection tools true to their claims?

RQ2. How do the AI-generated content detection tools perform when fed with original and AI-generated content in different ways on the same topic?

AI generated content detectors

This section introduces the AI-generated content detection tools that are used for the purpose of this study. Information about the AI detection accuracy of nine prevalent AI detection

Table 1.
Compilation of
different relevant
works and their
corresponding positive
and negative points

Reference	Area of analysis	Positive points	Negative points
Else (2023)	Scientific writing	Convincing scientific writing	Issue of academic integrity
Shoufan (2023)	Use of ChatGPT in computer engineering students	Assistance in student learning motivational study guide	Inappropriate responses to some queries Cannot replace human intelligence
Adamson (2023)	AI chatbots in warfare	Future technology	Risks of hidden mathematical models
Lo (2023)	Education	Human-like response	Varying performance for different subject domains
Neumann <i>et al.</i> (2023)	Higher education and scientific writing	Innovative Potential	Lack of rules and regulations for AI content
Ahmed and Sharo (2023)	Higher education	Better learning experiences	Negative effects on results
Dwivedi <i>et al.</i> (2023)	Multidisciplinary	Productivity gains	Plagiarism issues
Yu (2023)	Education	Integration of education with AI	Absence of thoughtful legislation
Crothers <i>et al.</i> (2023)	Machine text in general	Powerful open source model	Legal risks
Currie (2023)	Medical imaging education	Above average response quality for application of basic knowledge	Limitations of detection models
Homolak (2023)	Scientific writing	Human like time efficient response	Low quality responses for advanced analysis and design
Dalalah and Dalalah (2023)	Education and scientific writing	Logical and contextual responses	Ethical dilemmas such as acknowledgement of GPT as an author
Currie <i>et al.</i> (2023)	Nuclear medicine	AI integration capability for educating nuclear medicine	Inefficiency of AI content detection tools
Source(s): Authors' own work			False positives of AI content detection tools
			Capability for fabrication of knowledge

tools has been extracted, as it is claimed on the corresponding websites or information manual. Moreover, the minimum supported wordcount for the specified accuracy has also been identified. Further details, such as the AI model specifics and supported languages, have also been brought to the knowledge of the reader. A brief introduction of each of these tools is subsequently provided. Information about false positives (human written content flagged as AI generated) and false negatives (AI produced content flagged as human composed), where available, has also been provided.

Turnitin

The Turnitin AI detection model works by breaking the content into smaller chunks of data and passing them to an AI model trained for the content produced by GPT version 3.0 and 3.5. The name of the AI model is not specified by Turnitin, but it is trained on generative AI models which detect and suggest the next word in a sentence. Turnitin AI's model assigns a probability measure to each chunk based on how close the content of that particular chunk is to an AI-generated content. The mean of these probability measures is used to specify an overall score of the complete content. The model currently supports the English language

and a minimum wordcount of 300 words for best results. Turnitin also claims to have an accuracy of 98% and lower than one percent of false positives.

Originality

Originality is based upon supervised learning and incremental training of an AI model that is built upon multiple sub-models. The popular sub-model among these includes a Natural Language Processing (NLP)-based engine called Bidirectional Encoder Representation from Transformers (BERT). Claimed classification accuracy of Originality for AI content detection is 99% on GPT-4 and 83% on ChatGPT 4.

Sapling

Sapling's principle of checking original content starts with the calculation of probability for a single word in a sentence. Then the score is combined for a sentence and finally, accumulated for the complete content. Its principle is based on a transformer model, which is the same model used to predict and generate words in large language models. Its claimed accuracy is 97%, and false positives are less than 3%.

GPTZero

GPTZero utilizes a multi-layered deep learning model that checks the coherence of content with an AI-generated data, considering the fact that an AI-produced sentences follow a predicted pattern of words. It also analyses the content, sentence by sentence and further, examines its sentence structure from internet-wide data. It is trained on GPT version 4.0 as well. However, likewise, Turnitin, also works for well-structured English language only. Claimed classification accuracy of the GPTZero AI model is 99% percent for originally written articles and 85% for AI-produced content.

ZeroGPT.com

[ZeroGPT.com](https://www.zerogpt.com) has been trained on variety of training data through a complex deep learning model. It can also examine AI-generated text at a sentence level. Furthermore, it does not pose any limit on word count for AI detection to work appropriately. Contrary to most of the other AI content detectors, it also works for multiple languages, and its claimed accuracy is 98%.

Copyleaks

Precise details of the deep learning model behind Copyleaks are not known, however, it computes probabilistic scores for content to be AI generated. It works for fifteen different languages and its claimed accuracy is 99.1% in its latest release. CopyLeaks classifies the output into two categories, being human or AI generated and then, provides the likely percentage that the content was generated through the already labelled category.

HIVE

HIVE Moderation offers AI content detection not only for text but also for images and deepfakes. Their model works on the structure of how the content is formulated and returns probabilistic scores as to how likely the content is generated through an AI engine. They claim to have more than 99% accuracy of AI detection.

Smodin

Smodin AI detection supports multiple languages, and its detection model is based on sentiment analysis of text. In lenient mode, its accuracy is 99% for human written content and about 85% for AI written content. However, in strict mode, its accuracy is 95%.

Giant language model test room
Giant Language model Test Room (GLTR) provides a colour-segmented representation of detection based on three metrics. These metrics are based on word ranking among different predicted words, as it is done in the GPT2 model. Claimed accuracy is of GLTR is 72%.
A summary of the salient features of the aforementioned software is provided in [Table 2](#).

Materials and methods

The authors composed five different texts of 300 words or more, to be used as test samples, to meet the requirement of the minimum number of words of the above-mentioned tools. These five samples were written in the English language and treated as human originated content. These writings would serve as the first category of test data, called *Original*. Using this base material, another three categories of data were produced for each test sample, called *Paraphrased by AI*, *100% AI generated* and *100% AI generated with Contextual Information*. For the sake of brevity in the results section, test samples 1–5 are labelled as S1, S2, S3, S4 and S5, respectively, and the four categories of data against each test sample is called C1, C2, C3 and C4, respectively. No data have been needed or collected from any other human source; hence, any approvals were not needed. The authors curated the data themselves within the research parameters of the study.

The content for the “paraphrased by AI” category was produced by feeding the “original” articles, one at a time, to ChatGPT, which was then asked to rephrase the source text given in quotation marks. Next, the material for the category labelled “100% AI generated” was created by asking ChatGPT to produce 300 words on the said topic. Finally, for the last category, titled “100% AI generated with Contextual Information”, ChatGPT was given a specific context similar to the original text and asked to produce a 300-word response. The topics were selected from various domains of knowledge to provide a general idea of the performance of the AI content detection tools. The subjects for all the test samples were:

- (1) Indoor air quality index and its relation with mental health
- (2) Research efforts towards smart agriculture
- (3) Brain computer interface with respect to signal gathering techniques
- (4) Sustainable development goals
- (5) Reasons for lower intake in undergraduate electrical engineering programmes worldwide

Table 2.
Salient features of AI-generated content detection software

Sr. No.	AI detection tool	Model training	AI detection accuracy	Minimum word count
1.	Turnitin	GPT 3 and 3.5, ChatGPT	98%	300
2.	Originality	GPT 4, GPT 3, ChatGPT, Quillbot	99%	200
3.	Sapling	GPT-3.5 or ChatGPT	97%	50
4.	GPTZero	ChatGPT, GPT 4, BARD, Llama etc.	85%	250
5.	ZeroGPT.com	GPT 4, Chat GPT, BARD	98%	Not specified
6.	Copyleaks	GPT4, Chat GPT, BARD, Jasper, T5	99.1%	150 characters
7.	HIVE	GPT 3, ChatGPT	>99%	1,500 characters
8.	Smodin	ChatGPT, BARD	95–99%	Not specified
9.	GLTR	GPT 2, BERT	72%	Not specified

Source(s): Authors’ own work

These four versions of documents, against each sample, were then fed into these AI detectors and the corresponding AI scores were extracted. The mean scores of each individual tool and each category (Mean T) and the combined mean score of all tools for each category (Mean C1–Mean C4) are then calculated, which serve to benchmark the performance of one AI generated content detection tool against others as well as the overall performance of the tools. The version of ChatGPT used for producing content was 3.5.

Results

The results of this experiment in terms of AI content detection are given in Table 3. It must be mentioned here that CopyLeaks first labels the content as human or AI generated and then, adds a percentage of the likelihood of that content belonging to the already decided category. For such cases, we have assumed a simple complementary model to produce a number for the AI generated content. Moreover, GLTR AI Detector has not been included in the results as their model is older than ChatGPT 3.5. To be able to concisely present information in the Table 3, test samples are denoted as S1, S2, S3, S4 and S5, respectively, and the corresponding categories of datasets, as previously explained, are called C1, C2, C3 and C4, respectively. The objective of the research is not to highlight one tool's performance from the others or endorse/renounce any AI-generation content detection package, but gain insights into the performance of these tools from a research perspective. The eight tools are marked as Tool A (Turnitin), Tool B (Originality), Tool C (Sapling), Tool D (GPTZero), Tool E

		Tool A	Tool B	Tool C	Tool D	Tool E	Tool F	Tool G	Tool H
S1	C1	0%	0%	0.6%	0%	0%	27.9%	0%	0%
S2	C1	0%	0%	0%	0%	0%	2%	0%	0%
S3	C1	0%	0%	0%	0%	0%	3.4%	0%	30%
S4	C1	0%	0%	0.1%	0%	0%	1.2%	0%	0%
S5	C1	0%	0%	0%	0%	0%	3%	0%	0%
Mean T		0%	0%	0.14%	0%	0%	7.5%	0%	6%
Mean C1					1.71%				
S1	C2	0%	100%	0.4%	2%	36.11%	48.2%	0%	35.2%
S2	C2	0%	100%	0%	1%	0%	65.8%	0%	32.1%
S3	C2	0%	100%	0.01%	2%	0%	54.33%	0%	30.5%
S4	C2	0%	100%	100%	99%	8.92%	33.6%	0%	100%
S5	C2	0%	100%	100%	98%	36.96%	43.8%	50.9%	98%
Mean T		0%	100%	40.1%	40.4%	16.4%	49.1%	10.2%	59.2%
Mean C2					39.42%				
S1	C3	0%	100%	0%	4%	69.77%	71.3%	3.9%	82.2%
S2	C3	100%	100%	0%	51%	54.55%	83.4%	100%	62.5%
S3	C3	100%	100%	94.83%	32%	67.68%	79.4%	100%	81%
S4	C3	100%	100%	99.6%	99%	100%	88.9%	99.9%	100%
S5	C3	100%	100%	100%	78%	100%	74.7%	99.9%	97%
Mean T		80%	100%	58.9%	52.8%	78.4%	79.5%	80.7%	84.5%
Mean C3					76.85%				
S1	C4	92%	100%	0%	46%	46.64%	79%	100%	70.9%
S2	C4	0%	100%	0%	37%	37.45%	79.2%	0%	48.9%
S3	C4	0%	100%	4.19%	8%	17.44%	63.3%	6.4%	25.6%
S4	C4	0%	100%	100%	100%	100%	68.4%	99.9%	100%
S5	C4	0%	100%	99.7%	100%	97.83%	76.1%	99.9%	100%
Mean T		18.4%	100%	40.8%	58.2%	59.8%	73.2%	61.24%	69.1%
Mean C4					60.1%				

Source(s): Authors' own work

Table 3.
Percentage of
likelihood of AI-
generated content for
different sets of data

(ZeroGPT.com), Tool F (CopyLeaks), Tool G (HIVE) and Tool H (Smodin) in this and subsequent sections.

Comparative analysis of original content: C1

Original content created by a human has been consistently designated as “Human Content” by all the AI detectors for all three samples. Tool A, Tool B, Tool C, Tool D, Tool E and Tool G detected 0% or very close to 0% possibility of content being constructed by an AI engine. Tool F and Tool H produced a mean score of 7.5 and 6%, respectively, which also suggest that the content is most likely generated by a human. Based on these results, we can assume that AI detectors perform a good job of labelling human-originated content accurately.

Comparative analysis of paraphrased by AI content: C2

The content initially composed by humans and paraphrased by an AI engine has also been observed to be marked as human work by most of the AI detectors, with varying but small likelihood of being AI content except for Tool B and Tool H AI detection tools. Tool B AI detector noted this content to be AI produced content for all three samples with complete confidence, while Tool H had a mean score of 59.2%. This shows that at the current stage, most of the AI detectors were unsure of detecting any inappropriateness in the content.

Comparative analysis of 100% AI-generated content: C3

The content that was built wholly through AI engines seems to have been detected by the majority of the AI detectors. Around 6 of the 8 tools compared gave a mean score of more than 60% for the likelihood of the content being AI-generated. Tool C and Tool D, which could not predict to the levels of their counterparts, still presented a higher probability for the content to be AI-driven. Tool B, Tool E, Tool F and Tool H seem to have performed best as they consistently detected a higher probability of AI-generated content among the samples, while in the case of Tool A and Tool G, the results seemed to be hit or miss; which indicates that false negatives could be a problem with these tools.

Comparative analysis of 100% AI generated with contextual information content: C4

The content that is constructed through AI engines but with a lot of background and contextual information also seems to have perplexed the AI detection models as only two of the tools were able to predict with a probability higher than 70% that the content was generated through the use of an AI engine. All the remaining tools gave a mixed output; sometimes they missed out on finding the AI-produced material, and at other times they were completely sure that it was not naturally composed.

Cumulative analysis

If we look at the cumulative performance of the software on the whole, they seem to be doing an excellent job of detecting human-generated content (with the cumulative mean value of 1.71% AI content detected) and perform decently when detecting completely AI-generated content (with the cumulative mean value of 76.85% AI content detected). However, it must be said that they are not equally sure of their output when 100% AI-generated content is presented to them compared to completely human-produced text. Furthermore, they are confused while detecting paraphrased content (with the cumulative mean value of only 39.42% AI content detected) or the content that has been generated by providing contextual information (with the cumulative mean value of 60.1% AI content detected). This behaviour could potentially produce a lot of false negatives on one hand, which could be exploited by individuals with compromised academic integrity.

Discussion

Looking at the numbers, it can be easily deduced that the results of the AI-content detectors do not live up to the hype claimed in their sales pitch. The reason could be the lack of standardized test data or, in general, the nonexistence of any binding protocols in this domain. It must also be mentioned that this deficiency of any standardization could be rooted in the nascent nature of the industry, which is also rapidly evolving at the very same time. To exacerbate the problem, more and more platforms are recognizing this as a potential new market and launching new products/services quickly to get a portion of the business.

This state of affairs presents a real dilemma to the consumer, whether he/she is a researcher or an academic institute, when it comes to the employment and selection of a platform for detecting AI-generated content. Should they trust the tools they have already been using for plagiarism, as they have years of experience in the field of validating academic honesty? But clearly, the domain of detecting artificially generated content is new for them as well. Hence, their standing is probably almost similar to the new entrants. Since all the platforms are at an infant stage, should their claims and results produced be taken seriously and given any merit? Even when they are used, what should or should not be an acceptable threshold which determines if an article or thesis is original? What about the false positives? With plagiarism tools, the source from where the content is copied is also mentioned, which leaves no room for any false positives, but, in the scenario of AI-generated content detectors, where would the irrefutable proof come from that something was not original? This brings us back to our fundamental question: should the claims of such platforms be given any merit or should they even be employed at this stage in any formal way?

Another important question in this debate is about the ethical considerations of having AI-generated content for the purpose of learning and dissemination of knowledge. Where chatbots offer a very personalized learning experience and are lauded for their ability to provide a learning experience that is tailored to the learner, the same procedure of generating content becomes questionable when it is used to generate text that would be used to contribute to knowledge. The authors believe in the importance of academic integrity, but let's understand that these chatbots cannot produce new research like finding a cure to cancer or Alzheimer's disease, etc. They are just programmed to report the existing body of knowledge in a contextual way using very good authority of the language. So, if a researcher generates an article by giving a very specific setting, then the content generated in response may be AI-driven, but the actual set of conditions were thought of by the researcher and thus were original. So, the question arises whether the words used to deliver some content or the thoughts behind those words need to be original or both? Where many may argue both, in the authors' perspective, context should also be considered. To elucidate, uniqueness in the ideas/thoughts and the corresponding words that have been used to deliver those novel ideas should only be considered *together* in the fields where mastery of the language is to be exhibited, as in the fields of poetry or literature. In the scientific context, where a lot of research is produced by non-native speakers of the English language, more weight should be given to the originality of the work. At this point, the authors would point out that almost all the publishers provide a language editing service at some premium (Zakaria, 2022). Its purpose is to help improve the grammar and flow of language prior to publication, thereby improving the readability of the work that helps pave the path of getting it accepted for publication. So why should using an AI tool, which is probably free of cost, for precisely the purpose of improving the readability, be deemed any differently than using any such service. Such service providers may use a language expert to modify the manuscripts, so even the modified version is human-generated. But, before the age of the internet, search engines and online repositories, that bring us a plethora of articles in response to a few keywords in a matter of split seconds, it was equally tedious for any researcher to find relevant works to build their research on. So, should we accept these AI content generators as assistants which

could help improve the quality and presentation of research, or outlaw their use in the name of originality? With AI and data sciences creeping into every aspect of human life, a denial of their existence would be futile, as such tools are inevitably going to become part and parcel of everyday life.

However, their use comes with a caveat: using them to paraphrase someone's work and presenting it as one's own is not just unethical but blatant cheating. AI detectors should focus on such areas where they are able to prove rephrasing of someone's work so that the misuse of such tools could be eliminated.

Similarly, another dilemma that the AI-generated content detectors face is that many of them also provide services to generate AI-driven material. So, on one hand, they are generating content and detecting it on the other. The underlying principle of detecting artificial content has to be the same as generating it for authentic results. But if the detector algorithms know the generator algorithms, and they are more or less the same across the board, then eventually these companies are eating their own tails. And what would be the merit of their results in such scenarios?

Summarizing, the race for the detection of AI-generated content has just begun, and while it is unclear at this point which player has the right set of skills to emerge as the winner, the technology itself presents a dilemma for researchers, academic institutions and the detecting platforms themselves. There is a need to accept these tools and recognize the potential of their contribution in improving the existing body of knowledge and policies should be made to work in a beneficial way using these tools rather than scrapping them.

Conclusion

Although plagiarism detection tools are rapidly evolving and improving through feedback and training, there are still certain aspects that AI detection software lacks in handling. In general, they lack in detecting the source of AI-generated content. They are confused when presented with cultivated AI-generated data instead of simple AI-generated text. None of them live up to the hype of claimed accuracies. They are also based on AI models, which contain a certain bias towards producing results as human or AI-generated. Hence, considerable improvements are needed in AI models to validate the claims made by AI detection tools in the context to facilitate educators and researchers. Otherwise, there would always remain a question of integrity in the scientific articles and students' work submissions. There is also a dilemma where the same companies are striving to mature the AI detection capabilities of their tools while alongside developing engines to produce human like content. Keeping in view the current state of AI detectors, policymaking for ethical conduct in academic writing seems a big challenge for academic and research institutes. Similar research evaluating the performance of AI detection packages is a valuable tool in establishing academic honesty.

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